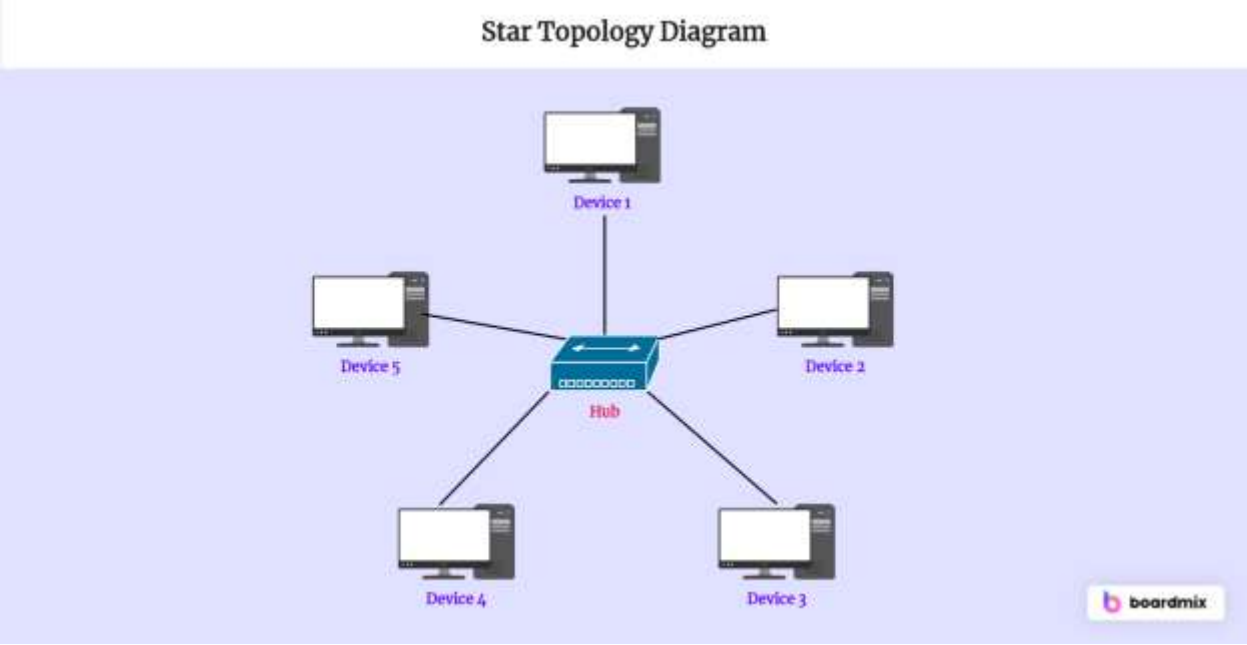
NETWORKTOPOLOGIES

TASK -1

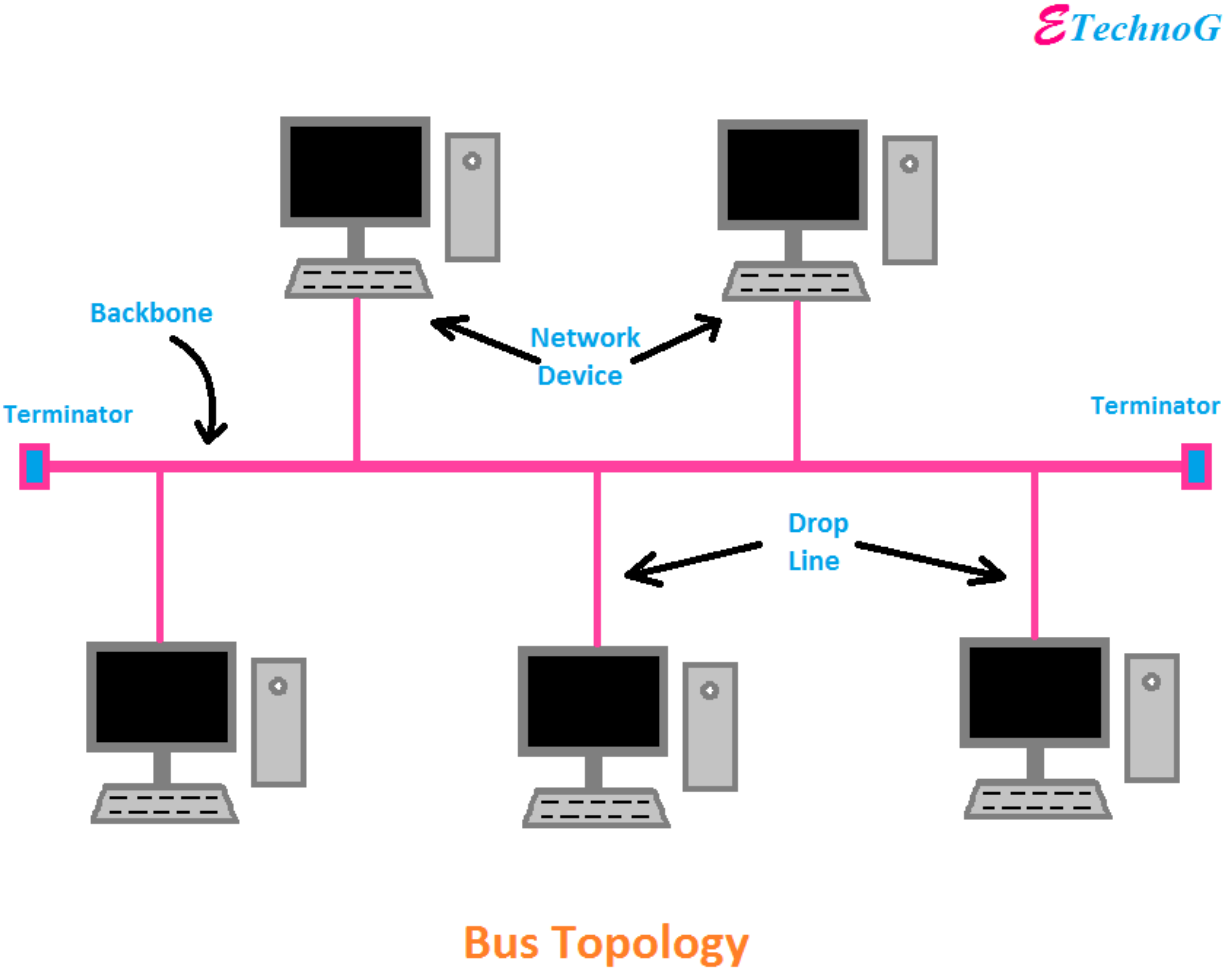
Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.



Real-world example: WhatsApp

Why: A central server can manage communication between many users. Each user connects to the central system to send and receive messages.

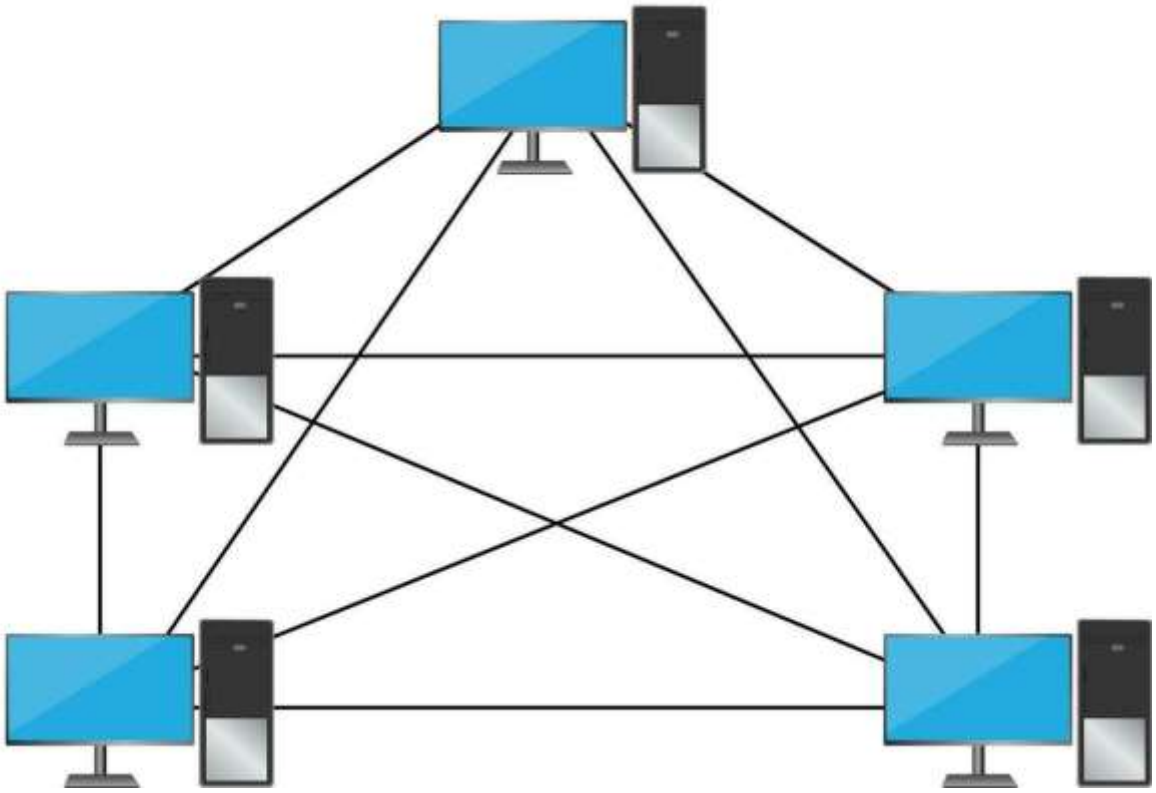
Bus Topology



Real-world example: Small office LAN

Why: Several computers can share one main communication cable, making it simple and relatively inexpensive for a small network.

Mesh topology



Real-world example: IPL streaming/CDN network

Why: Multiple network paths can connect different servers. If one path fails, traffic can use another path, improving reliability.

NETWORKTOPOLOGIES

TASK-2

Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Topology	Pros (Advantages)	Cons (Disadvantages)	Best Use Case	App Example
Star	1. Easy to install and manage 2. Easy to find faults 3. Failure of one device doesn't affect others 4. Easy to add/remove devices	1. Depends on central device 2. If switch/hub fails, whole network stops 3. Requires more cable 4. More expensive than bus	Offices, schools, homes, and LANs	**Instagram** – users connect to centralized servers/services
Bus	1. Simple design 2. Uses less cable 3. Low installation cost 4. Easy for small networks	1. Main cable failure affects the network 2. Performance decreases with more devices 3. Difficult to troubleshoot 4. Limited network length	Small and temporary networks	**Zomato** – a simple local network could connect devices in a small office

NETWORKTOPOLOGIES

Ring	1. Data travels in an organized manner 2. Less chance of data collision 3. Equal access for devices 4. Performs well under moderate traffic	1. Failure of one connection can affect the network 2. Difficult to troubleshoot 3. Adding/removing devices can be difficult 4. Slower if many devices are connected	Older LANs and specialized networks	**Paytm** – backend systems can use ring-like redundant network arrangements
Mesh	1. Very reliable 2. Multiple paths for data 3. Failure of one connection doesn't stop the network 4. Excellent fault tolerance	1. Expensive to install 2. Requires lots of cables/connections 3. Complex to configure 4. Difficult to maintain	Critical systems, data centers, wireless mesh networks	**Instagram** – distributed infrastructure can use multiple network paths for reliability

NETWORKTOPOLOGIES

Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.